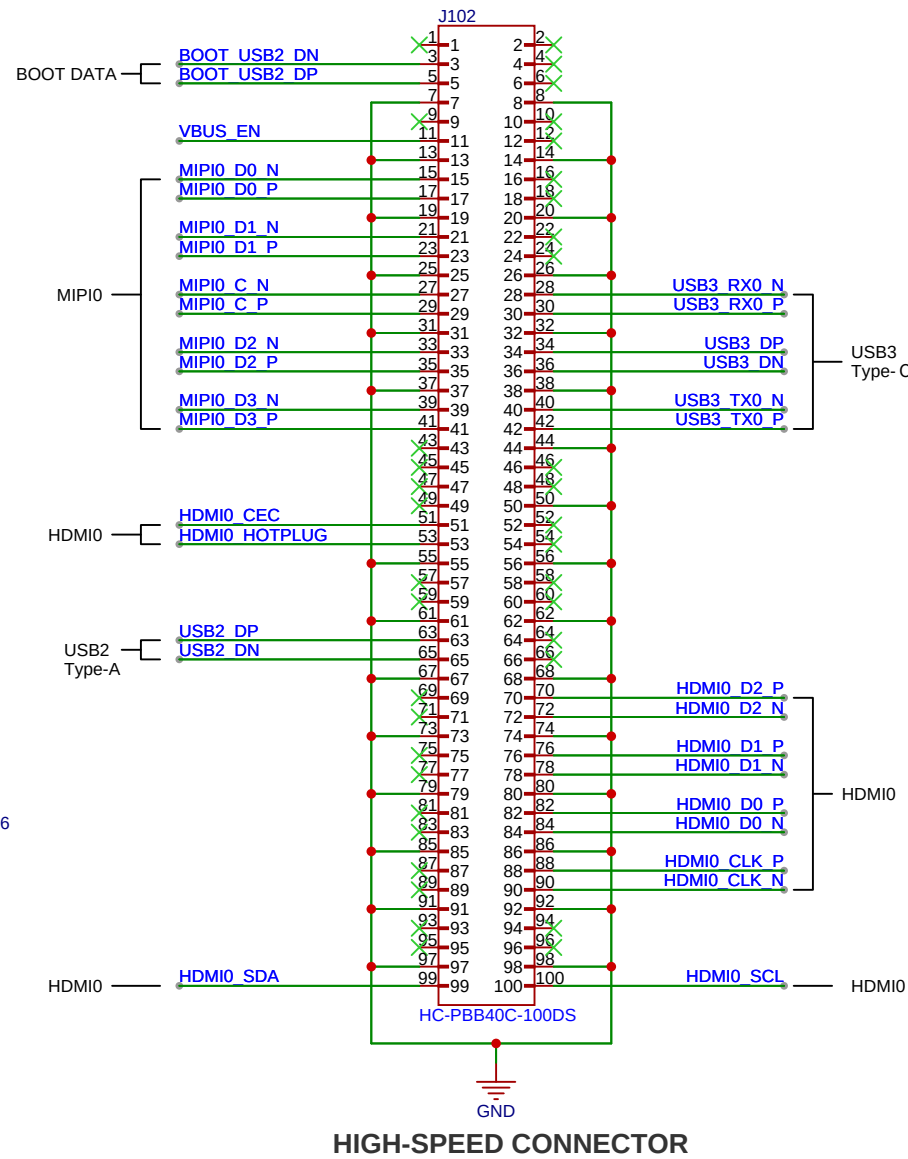
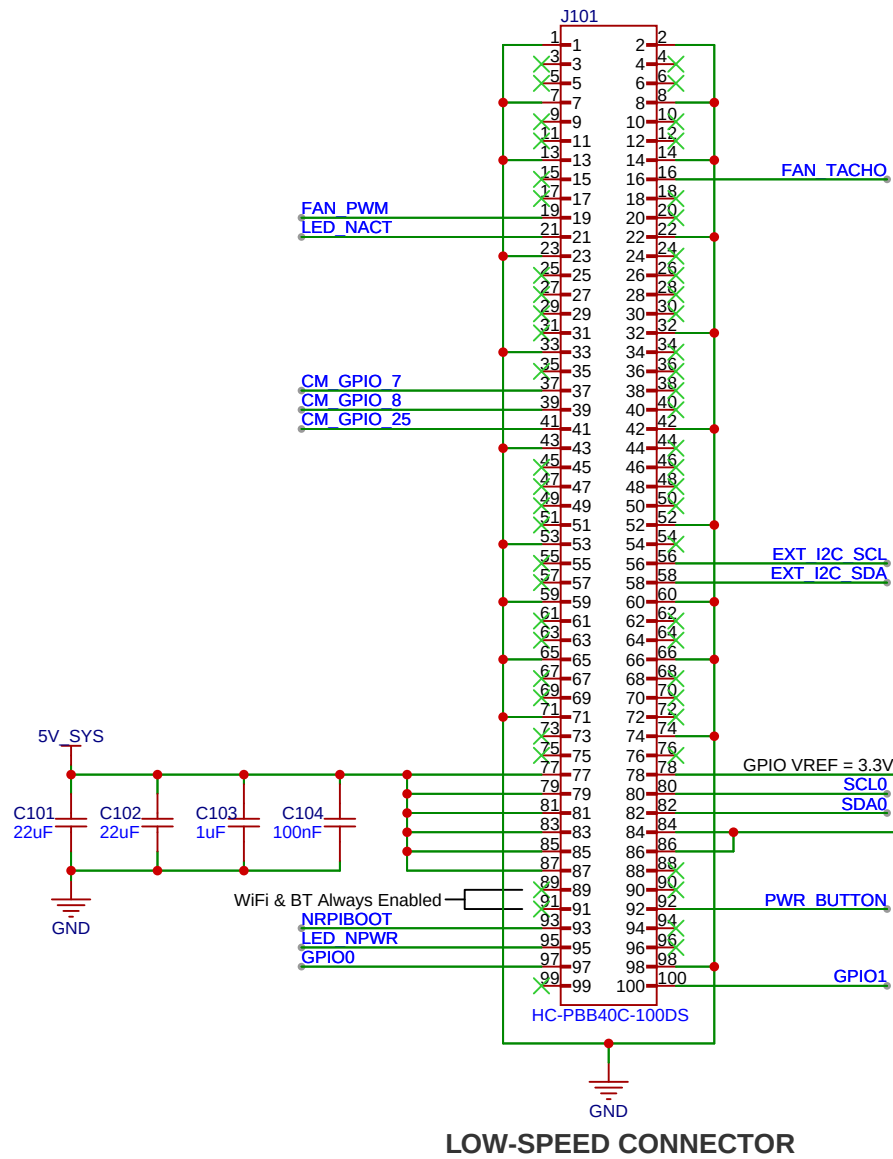


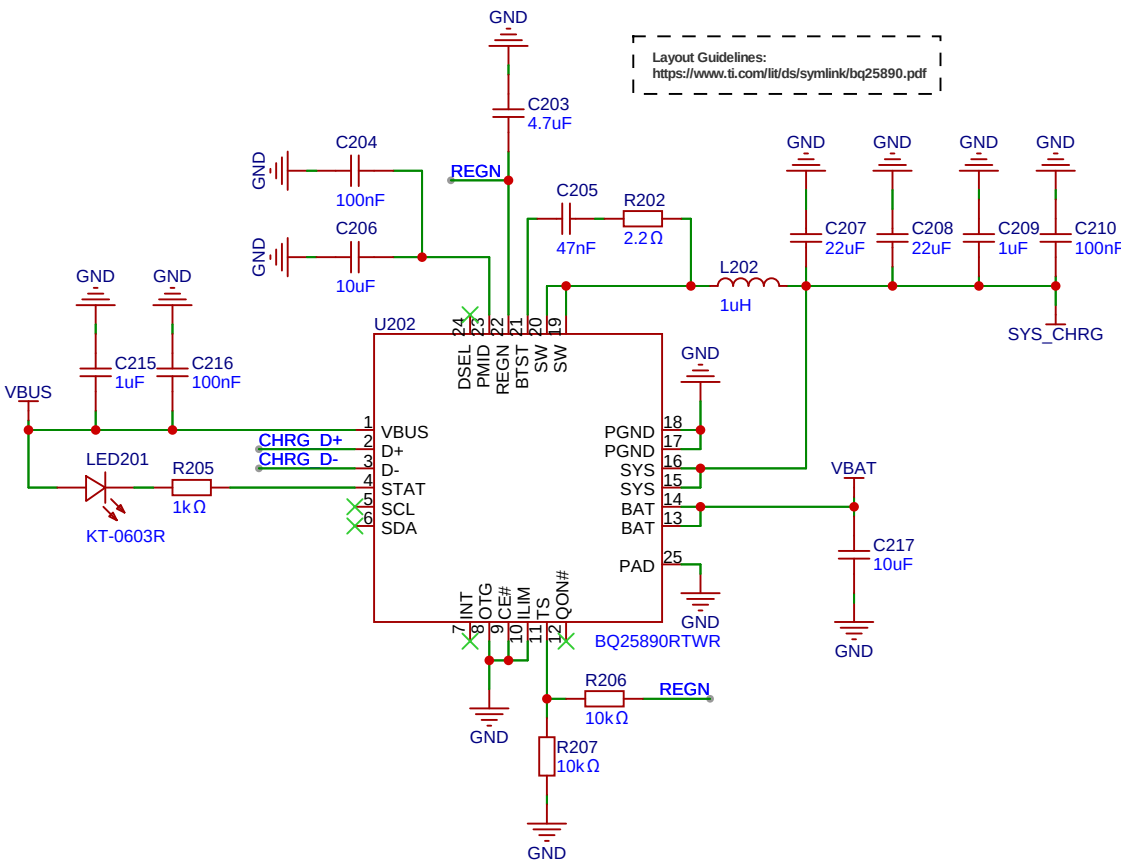
Raspberry Pi CM5 Pinout

Layout Guidelines:
<https://pip-assets.raspberrypi.com/categories/944-raspberry-pi-compute-module-5/documents/RP-008180-DS-6-cm5-datasheet.pdf>

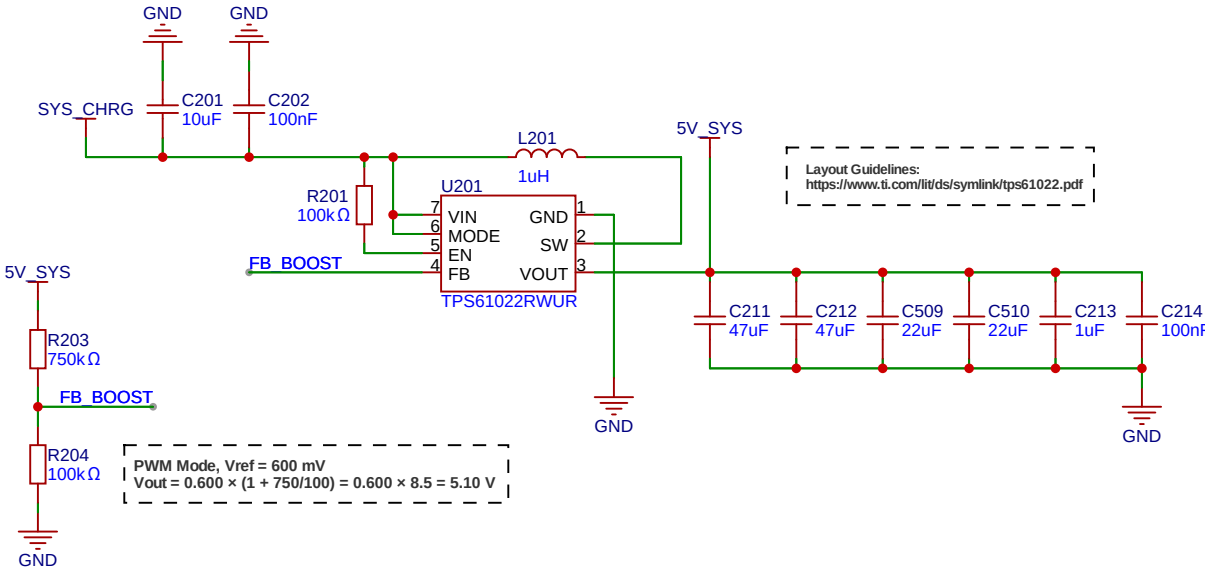


Schematic	SCH_MAINBOARD		Create at	2026-04-10
			Update at	2026-06-04
Board	MAINBOARD		Page	CM5 Module
Drawn	Horodnii O.	MIRAGE_3D_SCANNER		
Reviewed				
		Version	Size	Page 1 Total 5
EasyEDA		V1.1	A4	

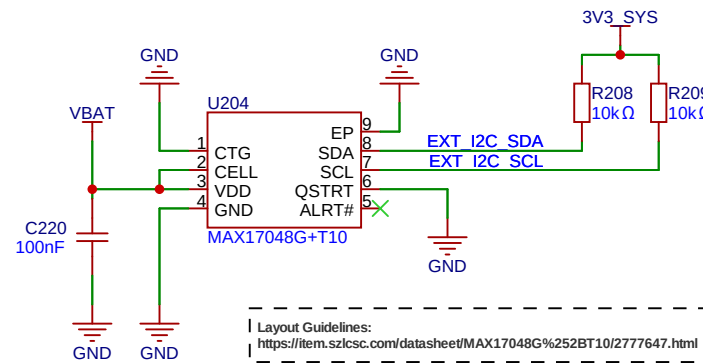
BATTERY CHARGER WITH POWER PATH



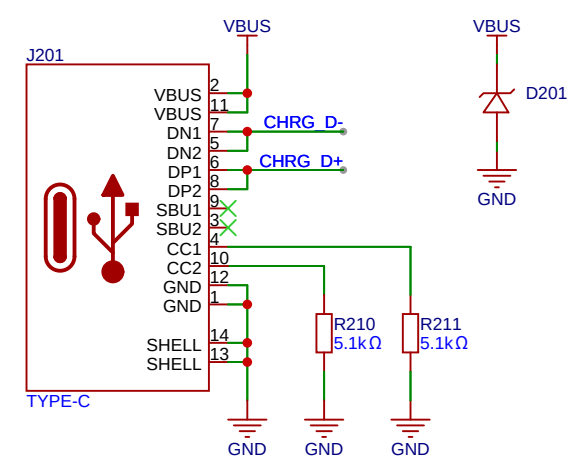
DC-DC BOOST



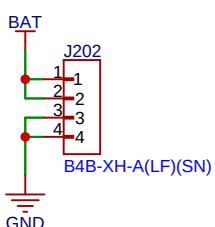
BATTERY FUEL GAUGE



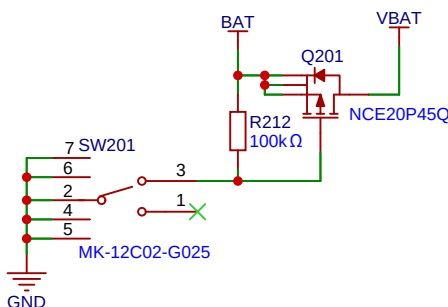
POWER CONNECTOR



BATTERY CONNECTOR



POWER SWITCH



Schematic	SCH_MAINBOARD			Create at	2026-04-10
				Update at	2026-06-04
Board	MAINBOARD			Page	Power Supply
Drawn	Horodnii O.	MIRAGE_3D_SCANNER			
Reviewed					
		Version	Size	Page 2 Total 5	
		V1.1	A4		

USB 2.0 Type-A (DEFAULT)

USB 2.0 Type-A (DEFAULT)

The diagram illustrates the electrical connections for a USB 2.0 Type-A connector. The connector (J301) is shown with its pins and the internal wiring to the USB 2.0 controller (U301) and the USB 2.0 load switch (U303).

Connector Pinout (J301):

- Pin 1: VCC (5V_USB2)
- Pin 2: D+ (USB2 DP)
- Pin 3: D- (USB2 DN)
- Pin 4: GND
- Pin 5: SHELL
- Pin 6: SHELL

Controller Pinout (U301):

- Pin 1: VCC (5V_USB2)
- Pin 2: D+ (USB2 DP)
- Pin 3: D- (USB2 DN)
- Pin 4: GND
- Pin 5: SHELL
- Pin 6: SHELL

Load Switch Pinout (U303):

- Pin 1: VBIAS (5V_USB2)
- Pin 2: VIN (5V_USB2)
- Pin 3: PG (GND)
- Pin 4: GND
- Pin 5: VOUT (5V_USB2)
- Pin 6: QOD (GND)

Components:

- U301: USB 2.0 Controller
- U303: USB 2.0 Load Switch (TPS2292SRXNR)
- D301: Diode
- C504: 22uF Capacitor
- C306: 1nF Capacitor
- C505: 22uF Capacitor
- C308: 100nF Capacitor

Layout Guidelines:

- Layout Guidelines: <https://www.ti.com/lit/ds/symlink/tps2292.pdf>

USB 2.0 Type-C (BOOT)

The schematic diagram illustrates the USB 2.0 Type-C (BOOT) circuit. It features a Type-C connector (J303) connected to a USB 2.0 controller (U305) and a USB 2.0 boot load switch (U310). The Type-C connector pins are labeled: 2 (VBUS), 3 (VBUS), 4 (DN1), 5 (DN2), 6 (DP1), 7 (DP2), 8 (SBU1), 9 (SBU2), 10 (CC1), 11 (CC2), 12 (GND), 13 (GND), 14 (SHELL), 15 (SHELL). The USB 2.0 controller (U305) pins are labeled: 1 (IO1), 2 (IO2), 3 (GND), 4 (IO3), 5 (IO4), 6 (NC), 7 (NC), 8 (GND), 9 (NC), 10 (NC). The USB 2.0 boot load switch (U310) pins are labeled: 1 (VBIAS), 2 (VIN), 3 (PG), 4 (GND), 5 (VOUT), 6 (QOD), 7 (ON), 8 (CT). The circuit includes a 5V_BOOT_USB2 supply, a 5V_SYS supply, and various passive components: R303 (5.1kΩ), R304 (5.1kΩ), C508 (22μF), C315 (1nF), C507 (22μF), and C317 (100nF). The diagram also shows a USB 2.0 boot load switch (U310) and a USB 2.0 boot load switch (U310).

USB 3.0 Type-C (DEFAULT)

USB 3.0 Type-C (DEFAULT)

AC Coupling Capacitors for TX is not needed, since it's already available inside CM5 module

Layout Guidelines:
<https://www.ti.com/lit/ds/symlink/hd3ss3212.pdf>

CM5 TX & RX P/N SWAPPED
CM5 Does Not Care About Differential Pair Polarity

TX1 & RX2 P/N SWAPPED
HD3SS3212 Does Not Care About Differential Pair Polarity

CC1 and CC2 signals swapped intentionally to invert DIR signal

PORT PULL UP = DFP MODE
CURRENT PULL UP = 900mA

USB 3.0 LOAD SWITCH

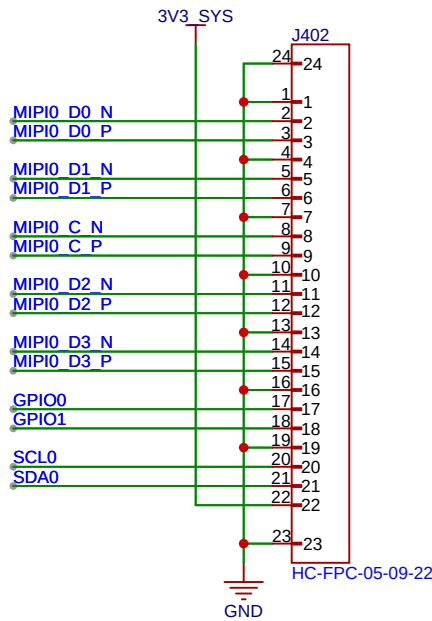
Layout Guidelines:
<https://www.ti.com/lit/ds/symlink/tps22992.pdf>

Layout Guidelines:
<https://www.ti.com/lit/ds/symlink/tusb321rwr.pdf>

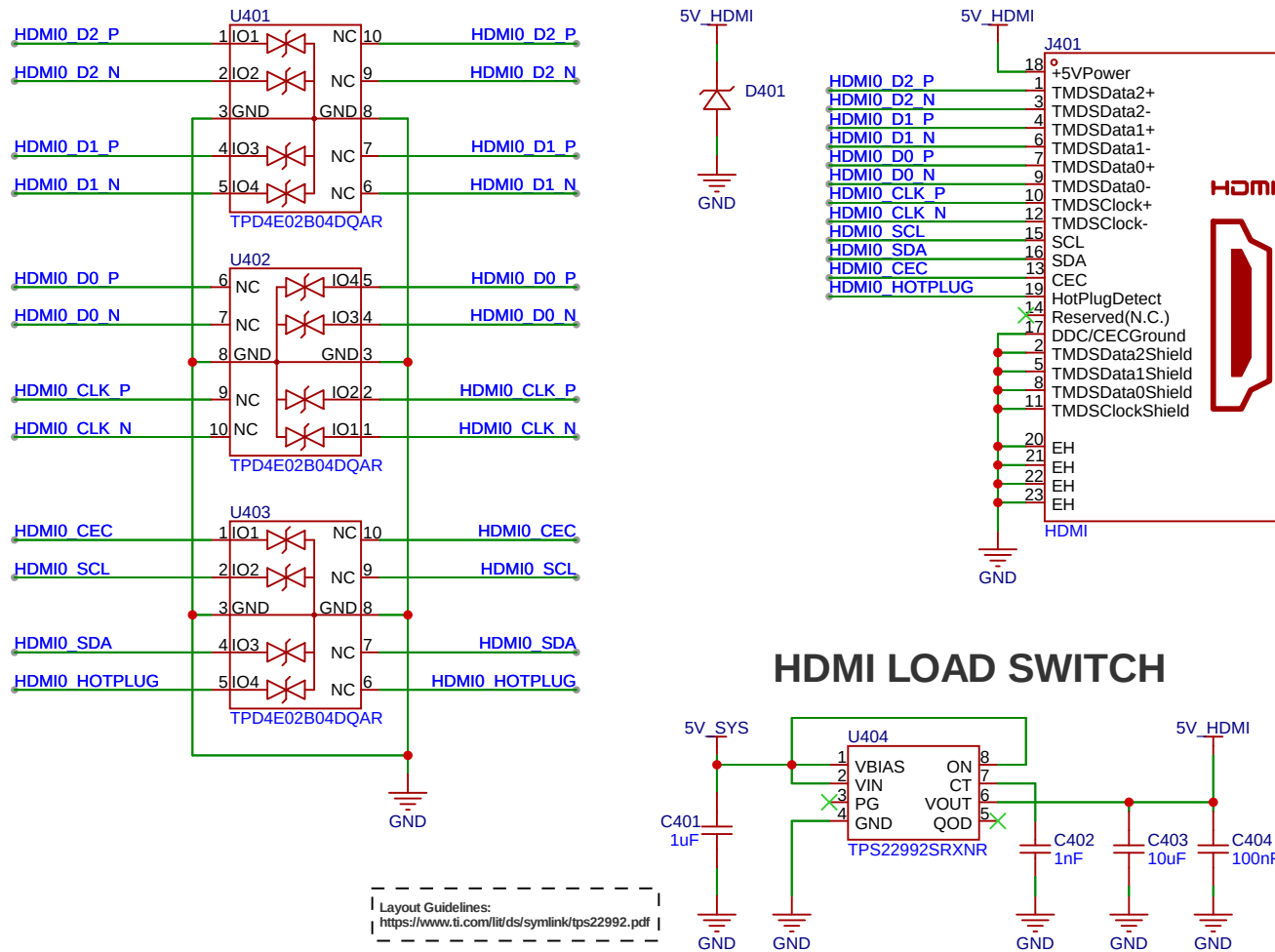
Layout Guidelines:
<https://www.ti.com/lit/ds/symlink/tpd4e02b04dqar.pdf>

Schematic	SCH_MAINBOARD			Create at	2026-04-10
Board	MAINBOARD			Update at	2026-06-04
Drawn	Horodnii O.	MIRAGE_3D_SCANNER			
Reviewed					
		Version	Size	Page 3 Total 5	
		V1.1	A4		

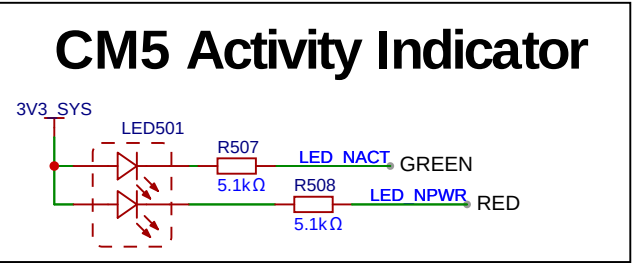
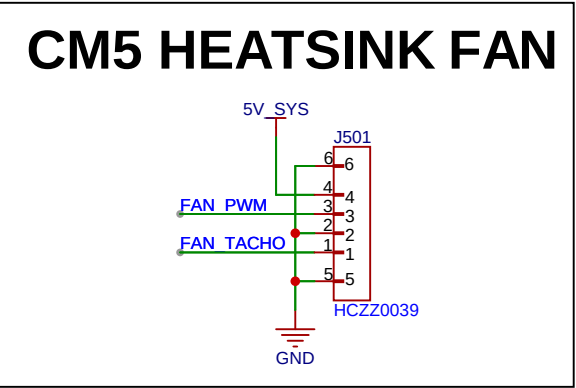
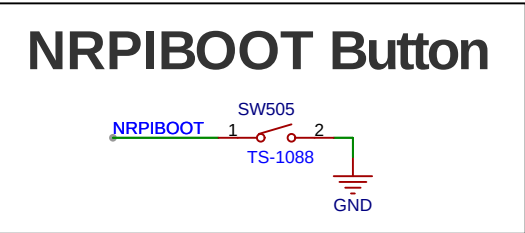
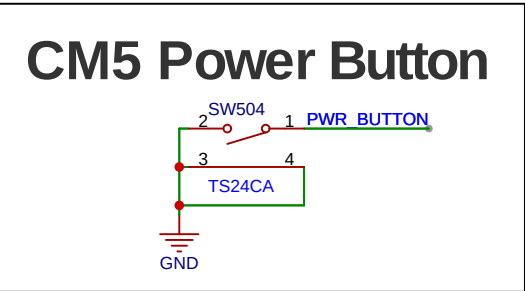
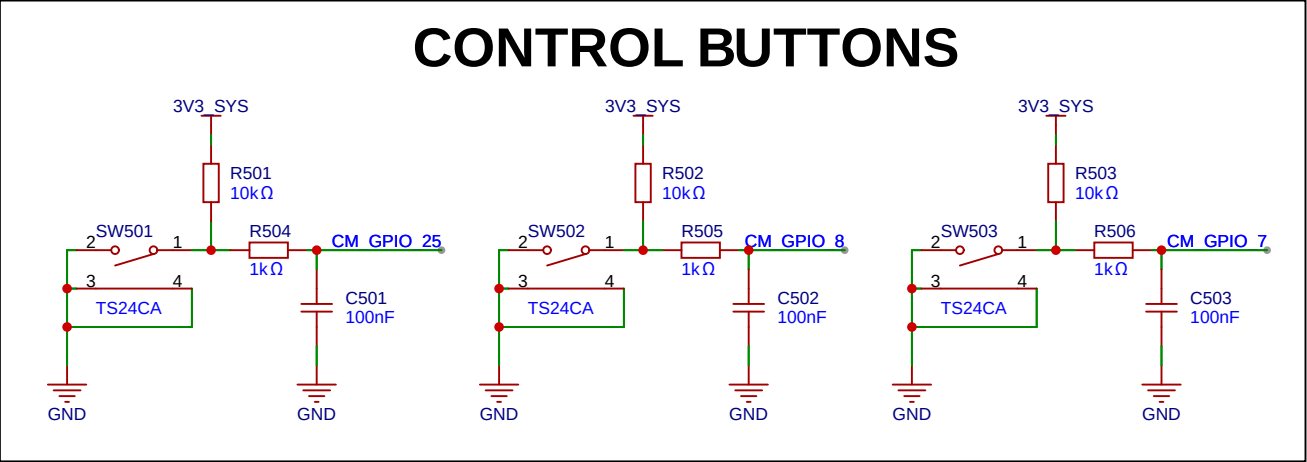
Display/Camera Connector



HDMI-A



Schematic	SCH_MAINBOARD			Create at	2026-04-10
				Update at	2026-06-04
Board	MAINBOARD			Page	Video and Display
Drawn	Horodnii O.	MIRAGE_3D_SCANNER			
Reviewed					
		Version	Size	Page 4 Total 5	
EasyEDA		V1.1	A4		



Schematic	SCH_MAINBOARD			Create at	2026-04-10
				Update at	2026-06-04
Board	MAINBOARD			Page	GPIO and Controls
Drawn	Horodnii O.	MIRAGE_3D_SCANNER			
Reviewed					
		Version	Size	Page 5 Total 5	
EasyEDA		V1.1	A4		